

Yash Vyas

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EDUCATION

Fergusson College (Autonomous), Savitribai Phule Pune University Pune, India
Bachelor of Science in Computer Science, CGPA: 8.55 / 10 (NEP 2020 Pattern) Aug. 2023 – May 2026

- **Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Relational Databases, Software Engineering, Statistics, Linear Algebra, Probability.
- **Self-Study:** Deep Learning, Reinforcement Learning, Computer Vision, LLM Fine-Tuning (SFT, GRPO, LoRA), Generative Models (CNNs, GANs), Medical Imaging.

EXPERIENCE

Sakal Media Group Pune, India
AI/ML Research Intern Jul. 2024 – Mar. 2026

- **Fine-tuned YOLOv8** on a custom newspaper-page dataset for layout-element detection (headlines, images, captions, ad blocks); built the labeling pipeline, augmentation strategy, and evaluation harness end-to-end.
- **Prototyped CNN** and **GAN** models for automatic newspaper-layout generation, exploring conditional generation conditioned on article length, image count, and template constraints.
- **Engineered** the data pipeline for the in-house **Automatic Layout System (ALS)**: parsing scripts for OCR-style text-region extraction, dataset versioning, and reproducible training runs.
- **Spearheaded** a 4-person ML R&D team (Dec '25 – Mar '26); proposed and secured GM/CTO approval for a new ML automation initiative through a formal research pitch.

PROJECTS

DesignGym 2.0 – RL Environment for LLM Layout Agents | *PyTorch, TRL, GRPO, Qwen, OpenEnv* Apr. 2026

- **Devised** an **OpenEnv-compliant Reinforcement Learning environment** that teaches an LLM to iteratively diagnose and improve layouts; **selected to Top 100 of 52,000+ developers** at the Meta × PyTorch × Hugging Face OpenEnv Hackathon Grand Finale, Bangalore. [\[HF Space\]](#) [\[Blog\]](#)
- **Computed** a 7-metric verifiable reward (overlap, alignment, spacing, balance, hierarchy, reading order, instruction fit) with **delta-aware, phase-aware reward shaping** to give GRPO a non-flat learning signal across long-horizon episodes.
- **Trained Qwen-0.5B** via heuristic-planner → **SFT (LoRA)** → **GRPO** pipeline; generated synthetic SFT data, ran ablations across reward configurations, and **lifted valid-action rate from 0% to 100%, instruction-fit by +8.5%, cut premature-finalize episodes by 83%**.
- Built the full evaluation harness (base vs. heuristic vs. SFT vs. GRPO) with deterministic seeded episodes; analyzed training curves and authored a research blog covering the math of computational aesthetics.

3D-Mind – Medical Imaging & Agentic AI for Brain Tumors | *nnUNet v2, NiBabel, SimpleITK, Gemini* Nov. 2025

- **Built** a browser-based platform that converts **MRI scans into interactive 3D brain visualizations** for remote radiology collaboration and tumor-progression tracking; **Top 100 of 20,000+ teams** at MumbaiHacks 2025, India's largest Agentic AI hackathon. [\[GitHub\]](#)
- **Integrated nnUNet v2** for 3D brain-tumor segmentation with **NiBabel & SimpleITK** for NIfTI medical-imaging I/O; designed inference services around a **Gemini**-powered agent that answers patient-history queries and generates clinical summaries on demand.

TECHNICAL SKILLS

Languages: Python, C++, SQL, Bash, JavaScript / TypeScript
ML / DL: PyTorch, Hugging Face Transformers, TRL (SFT, GRPO, LoRA / PEFT), scikit-learn, Reinforcement Learning, Reward Engineering, CNNs, GANs, Transformers, LLM Fine-Tuning, Model Evaluation & Ablation Studies
Computer Vision: YOLOv8, nnUNet v2, OpenCV, Edge Detection, OCR / ICR Pipelines, Medical Imaging (NIfTI, NiBabel, SimpleITK), 3D Volume Rendering
Data & Tools: Pandas, NumPy, Matplotlib, MySQL, SQLite, Jupyter, Google Colab, Hugging Face Hub & Spaces, Weights tracking, Git
ML Infra & Deployment: FastAPI, Docker, AWS, Linux, REST APIs, Gemini API, OpenAI-compatible inference
Other: Full-stack & cross-platform development (Next.js, React, Flutter / Dart, Node.js) – enables end-to-end ML product shipping.

HONORS & ACTIVITIES

Hackathons: **Top 100 of 52,000+ developers** – Meta × PyTorch × Hugging Face **OpenEnv Hackathon** Grand Finale, Bangalore (2026); **Top 100 of 20,000+ teams** – **MumbaiHacks 2025**, India's largest Agentic AI hackathon; **Runner-Up, Best Innovative Idea** – Persistent Smart School Hackathon (2020) for an embedded classroom-noise monitor at age 14.
Leadership & Activities: Founded the CS Department Newsletter at Fergusson College (2023 – 2026) – led student writing on AI & emerging tech; commit ~25 hrs/week to mentoring juniors on ML and college projects.